

Multimodal QA & GraphRAG

Roberto Giordano

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TOPICS:

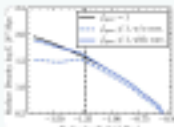
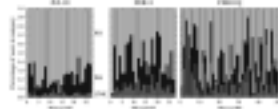
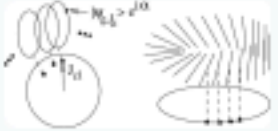
- **Multimodal QA**
 - Late interaction vs Late chunking
 - Visual retrieving vs Text retrieving
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Multigraph QA

Develop an analysis on late interaction vs late chunking strategies capable of integrating and processing information from scientific images to answer complex questions and select the correct choice.



DATASET

query string · <i>lengths</i> 	image image · <i>width (px)</i> 	image_filename string · <i>lengths</i> 	options string · <i>lengths</i> 	answer string · <i>classes</i> 
	243	22	33	67 values
Based on the graph, what is the impact of correcting for fspec not equal to 1 on the...		images/1810.10511_2.jpg	['A. Correction causes a significant increase in surface density across all...	C
Based on the progression from JUL10 to FEB11Q, what trend can be observed in the...		images/1107.3275_2.jpg	['A) A consistent decrease in the percentage of users with unknown...	D
What physical phenomenon could the pattern of lines on the right side of the figure...		images/quant-ph9912091_0.jpg	['A. Diffraction patterns of light', 'B. Magnetic field lines around a conductor', ...	B

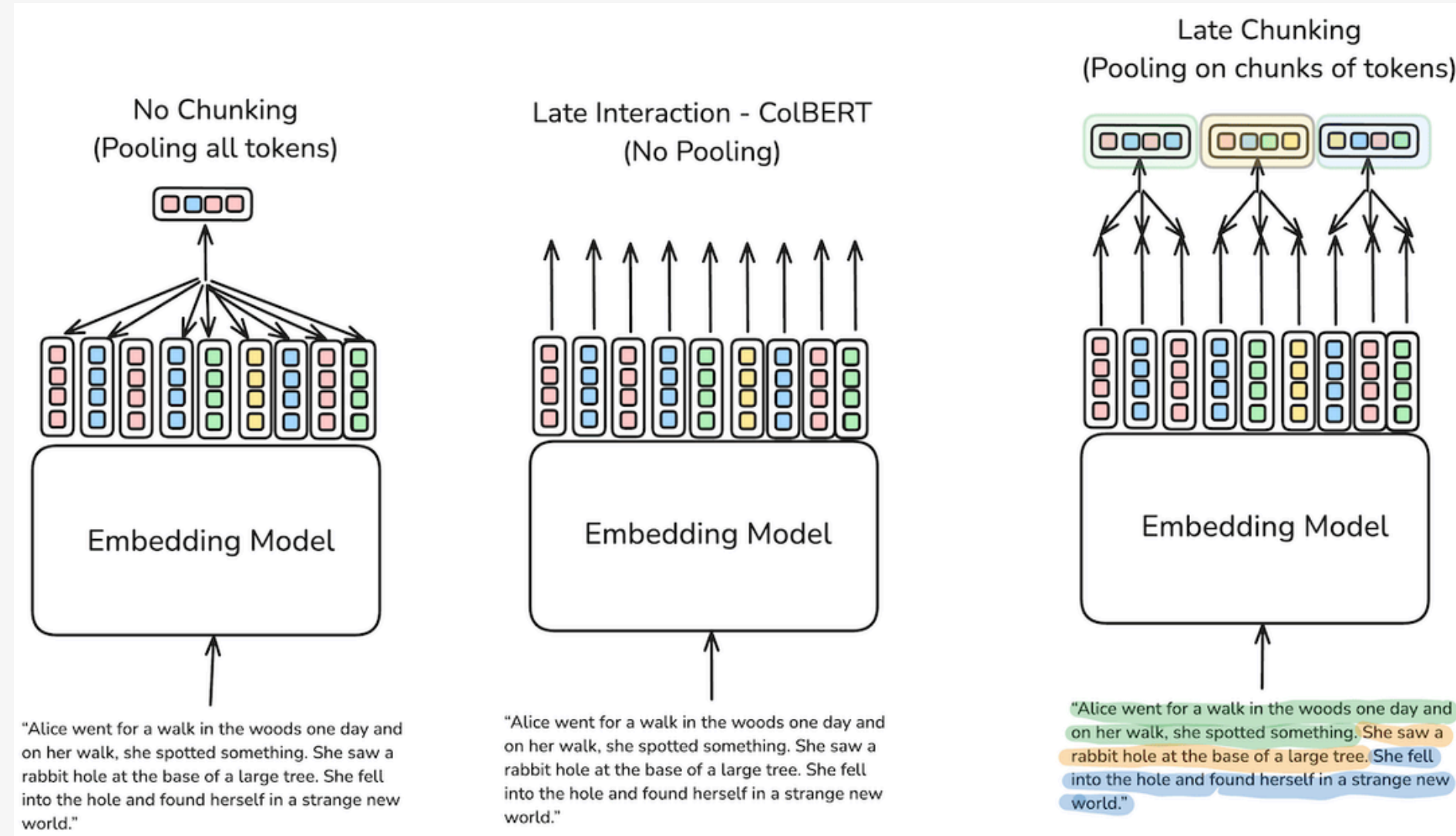
For each image, a description has been generated through Qwen2

Late Interaction

- Models like ColBERT process entire documents by generating embeddings for each token.
- During query interaction, token-level similarity is computed, enabling detailed comparisons at the cost of high computational and memory requirements

Late Chunking

- This approach embeds entire documents and divides them into chunks based on their embeddings.
- Each chunk retains global context, improving retrieval performance without excessive computational overhead



Models implemented

- 01 Visual late interaction: Colqwen2
- 02 Text late interaction: ColBERT
- 03 Text late chunking: Jina-embeddings

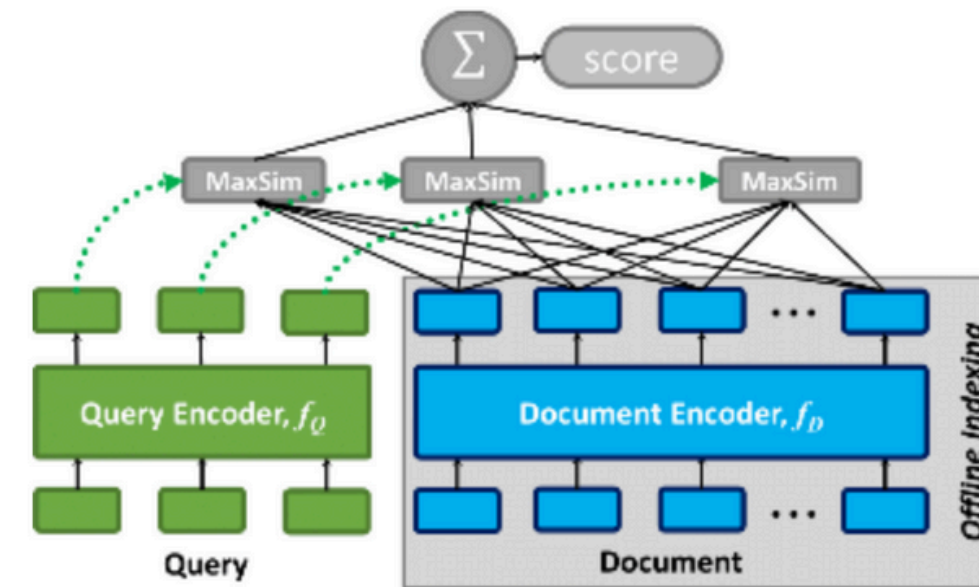
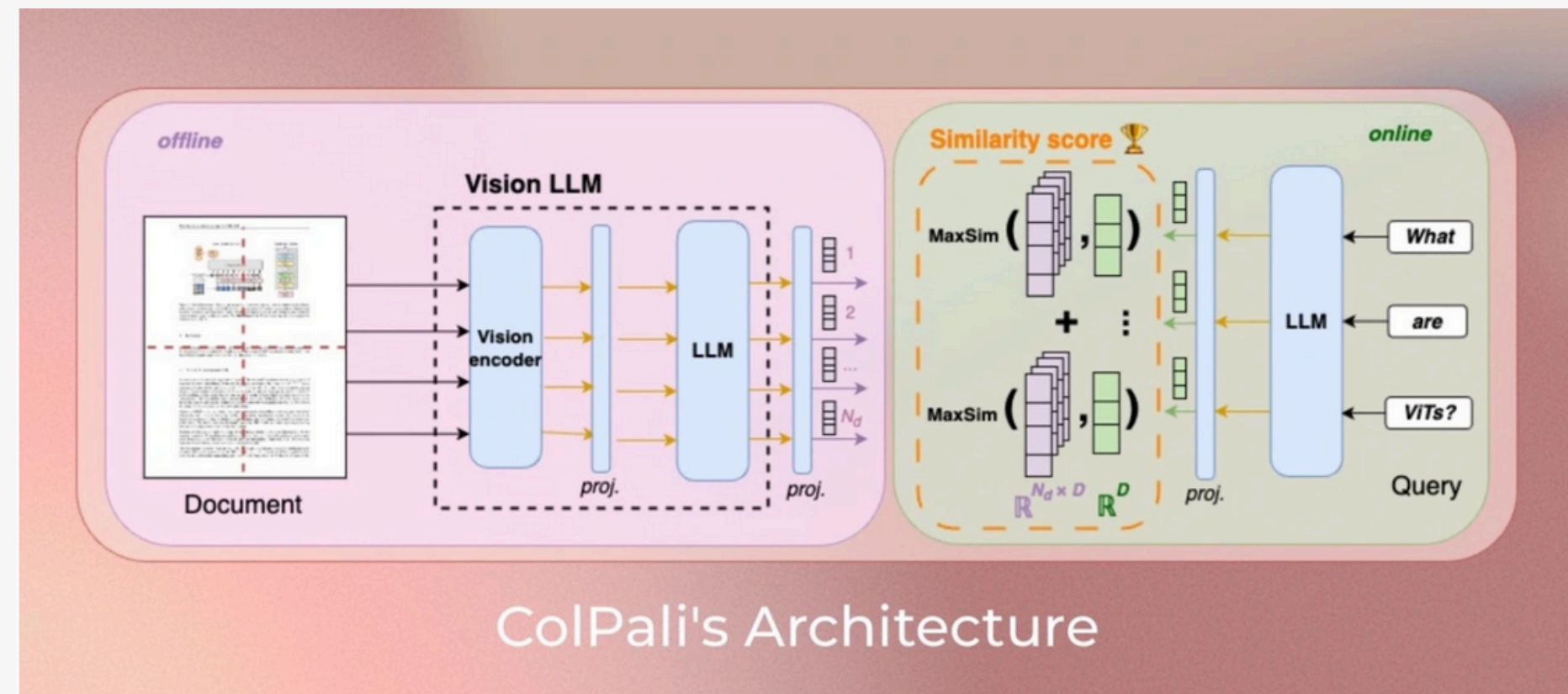


Figure 1: ColBERT's late interaction, efficiently scoring the fine-grained similarity between a queries and a passage.



Results on top-1 retrieval

Multimodal generative model: Qwen2-VL-2B-Instruct

	Colqwen (Visual late interaction)	ColBERT (Late interaction)	Jina (Late chunking)
Retrieval accuracy top-1	79.8%	47%	42.2%
Generation accuracy top-1	58.8%	49.79%	48.34%

This table illustrates the model's ability to retrieve the correct image (or text chunk) and, subsequently, to choose the correct answer from a multiple-choice list.

Results on top-3: retrieval accuracy

	Top-1	Top-2	Top-3
ColBERT (Late interaction)	47%	36.2%	19.4%
Jina (Late chunking)	42.2%	35.4%	28.8%

This table illustrates the retrieval correctness by evaluating the top-3 most similar embeddings, assessing whether each retrieved embedding is correct or not.

Results on top-3: generation accuracy

Multimodal generative model: Qwen2-VL-2B-Instruct

	Top-1	Top-2	Top-3
ColBERT (Late interaction)	50.64%	55.25%	54.64%
Jina (Late chunking)	50.24%	45.76%	47.22%

This table illustrates the generation correctness by evaluating the top-3 most similar embeddings, assessing whether each retrieved embedding is correct or not.

Results on top-3: retrieval accuracy

	At Least one correct	At Least two correct	All three correct
ColBERT (Late interaction)	55.6%	34.4%	12.6%
Jina (Late chunking)	47.8%	37.2%	21.4%

This table illustrates the retrieval correctness by evaluating the top-3 most similar embeddings, assessing the combination of retrieved embedding are correct or not.

Results on top-3: generation accuracy

Multimodal generative model: Qwen2-VL-2B-Instruct

	At Least one correct	At Least two correct	All three correct
ColBERT (Late interaction)	52.16%	54.65%	52.38%
Jina (Late chunking)	49.37%	46.77%	46.73%

This table illustrates the generation accuracy by evaluating the top-3 most similar embeddings, assessing the combination of retrieved embedding are correct or not.

GraphRAG

GraphRAG (Graph-based Retrieval-Augmented Generation) is particularly valuable for **multihop** questions, where answering requires reasoning across multiple pieces of interconnected information.

By structuring data as a graph, GraphRAG enables efficient navigation and aggregation of relevant **nodes** (e.g., text, tables, images), allowing the model to integrate diverse modalities and maintain the context of relationships.

PIPELINE

The steps underlined require an LLM call

01 Source documents

Subset of PDF Arxiv dataset from Vidore.

02 Text chunks

Divide the text in smaller chunks to extract better the entities. Trade-off needed.

03 Element instances

For each chunk, extract all the entities and relationships.

04 Element summaries

For each chunk, create a description of each entity and try to drop duplicates.

05 Graph Communities

After populating the graphDB, execute Leiden algorithm to cluster the entities.

06 Community summaries

For each community found, write a summary.

07 Community answers

For each summary, generate a partial answer to the question provided and let the LLM give a mark.

08 Global answers

Last call to the LLM that generate the final answer thanks to all the partial answers provided and sorted by mark.

Technologies and costs

Generated graph of 61
documents

Langchain
HuggingFace

Neo4j
Milvus

OpenAI: gpt-4o-mini
Ollama v 3.1

Document processing and integration costs ~0.06\$-0.08\$

Full communities (~600) answer generation: ~0.03\$



User's question: Which are the main themes of the documents?

The main themes of the documents encompass a diverse range of scientific fields and topics. Key areas of focus include:

1. **Nuclear Physics**: Research on charge radii and the structure of isotopes, highlighting collaborative efforts between institutions like Argonne National Laboratory (ANL) and GSI.
2. **Atomic Interactions**: The dynamics of ion-atom collisions, their impact on energy transfer, and the heating effects that are crucial for understanding atomic behavior.
3. **Astrophysics and Astronomy**: Exploration of exoplanets using advanced instruments such as the Gemini Planet Imager, along with challenges in imaging technology and astrometry. Additionally, the study of solar phenomena, including solar flares and their dynamics, is emphasized.
4. **Astrochemistry**: Investigation into the chemical processes in space, particularly the roles of water, methane, and carbon dioxide in the formation of stars and planetary systems.
5. **Bayesian Statistics**: The application of Bayesian nonparametric models, including the Dirichlet Process and Beta Distribution, focusing on their significance in probabilistic modeling and inference.
6. **Machine Learning**: The use of techniques like Latent Dirichlet Allocation (LDA) and its hierarchical variant (hLDA) in various scientific contexts, including the study of neutron star crusts and their thermal properties.

7. **Galactic Studies**: Analysis of galaxies, particularly the characteristics and evolution of quiescent versus star-forming galaxies, and the mechanisms that lead to quiescence.

8. **Molecular Biology**: Examination of motor proteins involved in cellular transport, the dynamics of muscle contraction, and the interaction between actin filaments and myosin motor proteins.

9. **Astrophysical Observations**: Measurement and analysis of beam sizes in observations, with a focus on Full Width at Half Maximum (FWHM) and the significance of molecular transitions in achieving effective angular resolution.

10. **Charm Physics**: The study of charm quarks, their mesons, decay processes, and the implications for new physics theories.

11. **Biological Dynamics**: The interaction of forces in biological processes, particularly in relation to molecular interactions and the dynamics of DNA replication forks.

12. **Computational Systems**: Simulation of two-processor systems, including interactions and decision-making processes within computational frameworks.

These themes reflect a rich tapestry of interdisciplinary research, showcasing advancements in understanding both fundamental and applied scientific questions across various domains.

Thank you!

Roberto Giordano